

Factor-Scale Localization of a Prime-Target Residual: Hard-Edge Bulk Persistence, Type-I Resolution, and Finite-Mellin Minimax Barriers

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Abstract

Fix $h \neq 0$ and a real-valued $W \in C_c^1((0, \infty))$. We attach to the prime-target residual the signed factor-scale measure

$$\nu_{X,h} = \sum_{m,n \geq 1} (\Lambda(n) - 1) \Lambda(mn + h) W(mn/X) \delta_{\log(n/m)}.$$

Its total mass, the zero Mellin mode, was evaluated previously by the identity $\sum_{n|r} (\Lambda(n) - 1) = \log r - \tau(r)$. The present paper asks how much localization in $\log(n/m)$ can be recovered rigorously.

There are two positive answers. Write

$$I_0(W) = \int_0^\infty W(u) du, \quad C_h = \prod_{p|h} (1 - p^{-1}) \prod_{p \nmid h} \left(1 + \frac{1}{p(p-1)}\right).$$

If both factors are restricted by $m, n > Y$, then, uniformly for $2 \leq Y \leq X^{1/3}$,

$$\sum_{m,n > Y} (\Lambda(n) - 1) \Lambda(mn + h) W(mn/X) = (1 - C_h) I_0(W) X \log X + O_{h,W}(X \log(2Y)).$$

The analogous $\Lambda(n)$ form has leading coefficient $I_0(W)$. Thus the complete leading coefficient survives after every fixed, polylogarithmic, or more generally $\log(2Y) = o(\log X)$ hard edge is removed from both factor coordinates. This is coarse bulk persistence, not a fixed-ratio Type II estimate. Second, maximal Bombieri–Vinogradov gives an asymptotic for every fixed smooth log-scale window in the Type I range. For $D \leq X^{1/2}(\log X)^{-B}$, it evaluates the insertion $V(\log(n/m) - s)$ uniformly in every center $s \in \mathbb{R}$.

We then give an exact boundary for deductions from finitely many Mellin samples or Taylor jets. On a total-variation ball, the minimax error for a window V equals its uniform distance from the exponential-polynomial space generated by the observed samples. Geometric divisor nodes $r = p^D$ produce explicit null profiles. At the prime-target row $3^4 + 2 = 83$, two adversarial, non-arithmetic probability profiles on genuine divisor nodes have identical jets through order three but different local window masses. On expanding log-scale intervals, any fixed finite sample/jet set that includes the zero-frequency value has limiting risk equal to the trivial midpoint risk. Resolving a fixed-width interior window from zero-frequency jets costs order $\log X$ jets at the information level.

Finally, the first Taylor interfaces are computed exactly. The first two nonconstant coefficients already contain convolutions of two von Mangoldt factors. A two-variable Mellin demodulation followed by an all-frequency sinc-squared kernel isolates the $m = 1$ layer exactly; its predicted asymptotic is precisely the corresponding single-weight fixed-shift Hardy–Littlewood assertion. No complete-tail localized asymptotic, fixed-shift prime-pair theorem, twin-prime lower bound, new distribution level, or breach of the parity barrier is claimed.

Keywords: prime-target residual; factor-scale localization; Bombieri–Vinogradov theorem; Selberg upper-bound sieve; Mellin data; optimal recovery; sieve parity.

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1 Introduction and claim ledger

The fixed-shift prime-pair problem asks for asymptotics of $\sum_n \Lambda(n)\Lambda(n+h)W(n/X)$ when h is fixed. For $h = 2$ this contains the twin-prime conjecture. Classical sieve and distribution theorems give powerful one-sided or averaged information, but they do not supply the required fixed-shift two-prime lower bound; see, for example, [4–7]. The purpose of this paper is narrower. We study which parts of a related factorized form can be localized using established theorems, and we quantify exactly what is not determined by finitely many Mellin coordinates.

Fix throughout an integer $h \neq 0$ and a real-valued function $W \in C_c^1((0, \infty))$. Terms for which $mn + h \leq 0$ are understood to vanish. Put

$$\nu_{X,h} = \sum_{m,n \geq 1} (\Lambda(n) - 1)\Lambda(mn + h)W(mn/X)\delta_{\log(n/m)}. \quad (1)$$

Its Fourier–Mellin transform is the finite exponential polynomial

$$\widehat{\nu}_{X,h}(t) = \int_{\mathbb{R}} e^{itu} d\nu_{X,h}(u) = \sum_{m,n \geq 1} (\Lambda(n) - 1)\Lambda(mn + h)W(mn/X)(n/m)^{it}. \quad (2)$$

For $V \in C_c^1(\mathbb{R})$ and a center $s \in \mathbb{R}$, the corresponding local factor-scale observation is

$$\mathcal{A}_{h,V}(X; s) = \int_{\mathbb{R}} V(u - s) d\nu_{X,h}(u). \quad (3)$$

The preceding paper [11] evaluated only the total mass $\widehat{\nu}_{X,h}(0)$. Write

$$I_0(W) = \int_0^\infty W(u) du, \quad I_1(W) = \int_0^\infty W(u) \log u du, \quad (4)$$

and

$$C_h(z) = \prod_{p|h} (1 - p^{-z-1}) \prod_{p \nmid h} \left(1 + \frac{1}{(p-1)p^{z+1}}\right), \quad C_h = C_h(0). \quad (5)$$

The exact divisor collapse

$$\sum_{d|r} (\Lambda(d) - 1) = \log r - \tau(r) \quad (6)$$

and the uniform Titchmarsh theorem of Assing, Blomer and Li [1] give, for every $A > 0$,

$$\begin{aligned} \widehat{\nu}_{X,h}(0) = X \Big[& (1 - C_h) \{(\log X) I_0(W) + I_1(W)\} \\ & - 2\{\gamma C_h + C'_h(0)\} I_0(W) \Big] + O_{A,h,W} \left(X (\log X)^{-A} \right). \end{aligned} \quad (7)$$

Complete factor aggregation is decisive in (6); it also destroys the location of the individual divisor d . The present paper separates three increasingly fine questions.

1.1 The positive results

The first result is a coarse but genuinely two-sided localization. If $m, n \leq Y$ cannot occur on the support of W —as is automatic for $Y \leq X^{1/3}$ and large X —then removing the two hard edges costs at most $O(X \log(2Y))$. Consequently the leading $X \log X$ coefficient in (7) remains in the interior $m, n > Y$ when $\log Y = o(\log X)$. The same statement holds for the unsigned prime-source form with coefficient 1; see [theorem 2.5](#). This does not confine n/m to a fixed dyadic window. It says that the leading aggregate is not carried by any fixed, polylogarithmic, or subpower collection of factor-edge layers.

The second result is fine localization on the directly accessible side. For

$$D \leq X^{1/2}(\log X)^{-B}, \quad (8)$$

maximal Bombieri–Vinogradov evaluates

$$\sum_{\substack{n \leq D \\ m \geq 1}} (\Lambda(n) - 1) \Lambda(mn + h) W(mn/X) V(\log(n/m) - s) \quad (9)$$

with an explicit main term, uniformly for every real center s ; see [theorem 3.1](#). Thus every fixed smooth log-scale window is available in the Type I range. The proof is an organization of the classical theorem, not an improvement of its distribution level.

1.2 The information boundary

A local window is an integral of the whole Mellin transform, rather than a function of finitely many frequencies. We make that distinction exact. For a total-variation ball of coefficient measures, arbitrary nonlinear reconstruction from finitely many samples and jets has minimax error

$$R \operatorname{dist}_{\infty} \left(V, \operatorname{span} \{ u^k e^{it_{\ell} u} : 0 \leq k < s_{\ell} \} \right). \quad (10)$$

This is the standard quotient-space principle of optimal recovery applied to the factor-scale coordinate; compare [9]. What is arithmetic here is the explicit realization of null directions on actual divisor geometries. The factor pairs of $r = p^D$ form an equally spaced log-scale grid, so a polynomial with prescribed unit-circle roots annihilates any specified finite set of samples and jets.

For $h = 2$, the row $r = 3^4 = 81$ has prime target 83. On its five factor nodes we exhibit two probability profiles having the same Mellin jets through order three but different central-window mass. This is an exact certificate on genuine divisor nodes and a genuine prime-target row, but the two profiles are deliberately adversarial: they are not the arithmetic weights $\Lambda(d) - 1$. The minimax theorem therefore rules out a deduction from finite Mellin data plus a norm bound alone. It does not assert that the actual prime residual realizes a worst-case direction.

1.3 The first arithmetic jets and the endpoint boundary

For a fixed product r , define

$$B_r(t) = \sum_{d|r} (\Lambda(d) - 1) e^{it(2 \log d - \log r)}.$$

The zero coefficient is (6). The next two coefficients already contain two-von-Mangoldt convolutions:

$$\begin{aligned} i^{-1} B'_r(0) &= (\log r)^2 - 2(\Lambda * \Lambda * \mathbf{1})(r), \\ i^{-2} B''_r(0) &= (\log r)^3 - Q_2(r) - 4((\mathfrak{D}\Lambda) * \Lambda * \mathbf{1})(r), \end{aligned}$$

where $(\mathfrak{D}f)(n) = f(n) \log n$ and Q_2 is explicit. Even symmetrization removes the odd coefficient, but not the second one. These are exact identities, not cancellation estimates.

At the opposite extreme, an all-frequency kernel after a two-variable Mellin demodulation selects the $m = 1$ layer exactly. Its evaluation with the Hardy–Littlewood main term is therefore equivalent to the weighted fixed-shift prime-pair asymptotic, rather than a proof of it.

1.4 One-way claim ledger

For clarity, the logical content is summarized in [table 1](#).

Input	Proved output	Not obtained
Complete zero mode + upper sieves	Bulk persistence after $m, n \leq Y$ are deleted	Fixed-ratio or signed Type II cancellation
Maximal Bombieri–Vinogradov	Smooth scale localization for $n \leq D$	Localization of the complete $n > D$ tail
Finite Mellin samples/jets + a TV bound	Exact minimax radius and null profiles	Impossibility for the actual $\Lambda - 1$ profile
Taylor and Fourier identities	Explicit parity-sensitive interfaces and exact $m = 1$ selector	Hardy–Littlewood asymptotic or twin-prime lower bound

Table 1: Every implication in the paper is one-way.

2 Hard-edge removal and bulk persistence

We begin with a coarse form of scale localization. It removes both factor edges but leaves the ratio n/m otherwise unrestricted. Define

$$\mathcal{P}_h(X; W) = \sum_{m, n \geq 1} \Lambda(n) \Lambda(mn + h) W(mn/X), \quad (11)$$

$$\mathcal{A}_h(X; W) = \sum_{m, n \geq 1} (\Lambda(n) - 1) \Lambda(mn + h) W(mn/X), \quad (12)$$

and let a superscript $\text{bulk}(Y)$ mean that both $m > Y$ and $n > Y$ are imposed.

2.1 The two complete collapses

The prime-source form has an even simpler zero-mode collapse than the residual form.

Proposition 2.1 (Complete prime-source collapse). *For every $A > 0$,*

$$\mathcal{P}_h(X; W) = X \{ (\log X) I_0(W) + I_1(W) \} + O_{A, h, W} \left(X (\log X)^{-A} \right). \quad (13)$$

Proof. Put $r = mn$ and use $\sum_{n|r} \Lambda(n) = \log r$. Then

$$\mathcal{P}_h(X; W) = \sum_{r \geq 1} (\log r) \Lambda(r + h) W(r/X).$$

The fixed-shift prime number theorem and smooth partial summation give

$$\sum_r (\log r) \Lambda(r + h) W(r/X) = \int_0^\infty (\log u) W(u/X) du + O_{A, h, W} (X (\log X)^{-A}),$$

and the integral is the main term in (13). □

The residual form is (7). Only its leading coefficient is needed below:

$$\mathcal{A}_h(X; W) = (1 - C_h) I_0(W) X \log X + O_{h, W}(X). \quad (14)$$

2.2 An edge ledger

We record the upper bounds used to remove the two factor edges. They are upper-sieve statements; no cancellation between rows is asserted.

Lemma 2.2 (One-prime edge). *Uniformly for $2 \leq Y \leq X^{1/3}$,*

$$\sum_{q \leq Y} (\Lambda(q) + 1) \sum_{k \geq 1} \Lambda(qk + h) |W(qk/X)| \ll_{h,W} X \log(2Y), \quad (15)$$

$$\sum_{q \leq Y} \sum_{k \geq 1} \Lambda(qk + h) |W(qk/X)| \ll_{h,W} X \log(2Y). \quad (16)$$

Proof. For $(q, h) = 1$, the weighted Brun–Titchmarsh inequality and partial summation give

$$\sum_{k \geq 1} \Lambda(qk + h) |W(qk/X)| \ll_{h,W} \frac{X}{\varphi(q)} \quad (17)$$

uniformly for $q \leq X^{1/3}$. The proper prime-power targets contribute at most $O_W(X^{1/2}(\log X)^2)$ in one row, which is $O(X/\varphi(q))$ in this range for large X : there are $O_W(X^{1/2})$ powers $p^a \asymp_W X$ with $a \geq 2$, and each von Mangoldt weight is $O(\log X)$. The elementary estimates

$$\sum_{q \leq Y} \frac{1}{\varphi(q)} \ll \log(2Y), \quad \sum_{q \leq Y} \frac{\Lambda(q)}{\varphi(q)} \ll \log(2Y) \quad (18)$$

then prove the coprime part of both claims.

If $(q, h) > 1$ and $qk + h$ is in the support of Λ , some prime $p \mid h$ divides $qk + h$. Hence $qk + h = p^a$. For each of the finitely many $p \mid h$, only $O_{h,W}(1)$ powers lie in the fixed-ratio interval forced by $W(qk/X)$. Summing the divisor weights over the corresponding $q \mid p^a - h$ costs X^ε for every $\varepsilon > 0$, which is absorbed by the right-hand sides. $\square$

The other edge contains two prime weights. Its standard Selberg-sieve bound is included with the local-factor calculation because that dependence on the varying coefficient m is important.

Lemma 2.3 (Two-linear-forms edge). *Uniformly for $2 \leq Y \leq X^{1/3}$,*

$$\sum_{m \leq Y} \sum_{n \geq 1} \Lambda(n) \Lambda(mn + h) |W(mn/X)| \ll_{h,W} X \log(2Y). \quad (19)$$

Proof. First suppose $(m, h) = 1$ and restrict both von Mangoldt factors to primes. On an interval $n \asymp_W X/m$, apply the Selberg upper-bound sieve to the two forms

$$n, \quad mn + h.$$

If ν_p is the number of roots of $n(mn + h)$ modulo p , then

$$\nu_p = \begin{cases} 1, & p \mid mh, \\ 2, & p \nmid mh. \end{cases}$$

If m and h are both odd, then n and $mn + h$ cannot both be odd primes in the ranges $n \asymp_W X/m$ and $mn + h \asymp_W X$ once X is large; the prime–prime contribution is then zero. In the admissible cases the upper singular factor satisfies

$$\mathcal{S}^+(m, h) \ll_h \prod_{\substack{p \mid m \\ p > 2}} \frac{p-1}{p-2} \ll_h \frac{m}{\varphi(m)}. \quad (20)$$

Indeed, after comparison with $p/(p-1)$ the quotient of the two local factors is $1 + O(p^{-2})$. The standard dimension-two upper sieve [7, Chapter 6] therefore gives

$$\sum_{\substack{n \geq 1 \\ n, mn+h \text{ prime}}} (\log n) \log(mn+h) |W(mn/X)| \ll_{h,W} \frac{X}{\varphi(m)}. \quad (21)$$

Summing (21) and using (18) costs $O_{h,W}(X \log(2Y))$.

It remains to restore proper prime powers. If n is a proper prime power, their aggregate contribution is bounded by

$$\sum_{m \leq Y} (X/m)^{1/2} (\log X)^2 \ll \sqrt{XY} (\log X)^2.$$

If $mn+h$ is a proper prime power, the crude aggregate bound

$$\sum_{m \leq Y} X^{1/2} (\log X)^2 \ll Y X^{1/2} (\log X)^2$$

suffices. Both are $o(X)$ in the stated range. Finally, if $(m, h) > 1$ and $\Lambda(mn+h) \neq 0$, the same argument as in lemma 2.2 forces $mn+h = p^a$ for a fixed prime $p \mid h$; the total is $X^{o(1)}$. This proves the lemma. $\square$

Remark 2.4 (Why the power range is deliberately conservative). The exponent $1/3$ is not optimized. It makes all proper-prime-power ledgers $o(X)$ with no need for a refined progression count. The result below concerns subpower hard edges, for which this range is more than sufficient.

2.3 The bulk theorem

Theorem 2.5 (Two-sided hard-edge bulk persistence). *Fix $h \neq 0$ and $W \in C_c^1((0, \infty))$. Uniformly for*

$$2 \leq Y \leq X^{1/3}, \quad (22)$$

one has

$$\mathcal{P}_h^{\text{bulk}(Y)}(X; W) = I_0(W) X \log X + O_{h,W}(X \log(2Y)), \quad (23)$$

$$\mathcal{A}_h^{\text{bulk}(Y)}(X; W) = (1 - C_h) I_0(W) X \log X + O_{h,W}(X \log(2Y)). \quad (24)$$

Proof. Choose $0 < \alpha < \beta$ with $\text{supp } W \subset [\alpha, \beta]$. In the range (22), the intersection $m, n \leq Y$ is disjoint from $mn \asymp_W X$ once X is sufficiently large. Thus each bulk form is its complete form minus the $n \leq Y$ edge minus the $m \leq Y$ edge.

For the prime-source form, the n -edge is bounded by the $\Lambda(q)$ part of (15); the m -edge is lemma 2.3. Insert (13).

For the residual form, bound $|\Lambda(n) - 1| \leq \Lambda(n) + 1$. The n -edge follows from (15). On the m -edge, the $\Lambda(n)$ part is lemma 2.3, and the constant part is (16). Insert (14). All discarded $O(X)$ terms are absorbed by $O(X \log(2Y))$. $\square$

Corollary 2.6 (Subpower edge removal). *Suppose $2 \leq Y = Y(X) \leq X^{1/3}$ and $\log(2Y) = o(\log X)$. If $I_0(W) \neq 0$, then*

$$\mathcal{P}_h^{\text{bulk}(Y)}(X; W) \sim I_0(W) X \log X.$$

If additionally $(1 - C_h) I_0(W) \neq 0$, then

$$\mathcal{A}_h^{\text{bulk}(Y)}(X; W) \sim (1 - C_h) I_0(W) X \log X.$$

In particular, the conclusion applies to $Y = (\log X)^K$ and to $Y = \exp((\log X)^\alpha)$ for fixed $K > 0$ and $0 < \alpha < 1$.

Remark 2.7 (What persists and what does not). When $Y = X^\delta$, the error in [theorem 2.5](#) has the same order as the displayed main term; no unchanged coefficient is then asserted. Even in the subpower range, the theorem controls a wide interior aggregate, not a fixed-width window in $\log(n/m)$. It is therefore neither a Type II estimate nor a prime-pair asymptotic.

3 Fixed smooth scale windows in the Type-I range

The bulk theorem removes only the two outer hard edges. We now insert an arbitrary fixed smooth window in the logarithmic ratio, at the cost of restricting the source factor n to the classical Type I range.

For $V \in C_c^1(\mathbb{R})$, $s \in \mathbb{R}$ and $D \geq 1$, define

$$\mathcal{A}_{h,\leq D,V}(X; s) = \sum_{\substack{n \leq D \\ m \geq 1}} (\Lambda(n) - 1) \Lambda(mn + h) W(mn/X) V\left(\log \frac{n}{m} - s\right). \quad (25)$$

Theorem 3.1 (Type-I factor-scale localization). *Fix $h \neq 0$, $W \in C_c^1((0, \infty))$, $V \in C_c^1(\mathbb{R})$ and $A > 0$. There is $B = B(A, h, W, V) > 0$ such that, uniformly for every $s \in \mathbb{R}$ and*

$$1 \leq D \leq \frac{X^{1/2}}{(\log X)^B}, \quad (26)$$

one has

$$\begin{aligned} \mathcal{A}_{h,\leq D,V}(X; s) &= X \sum_{\substack{n \leq D \\ (n,h)=1}} \frac{\Lambda(n) - 1}{\varphi(n)} \int_0^\infty W(u) V\left(\log \frac{n^2}{Xu} - s\right) du \\ &\quad + O_{A,h,W,V}\left(\frac{X}{(\log X)^A}\right). \end{aligned} \quad (27)$$

Proof. First suppose $(n, h) = 1$ and put $t = nm + h$. The target prime lies in the reduced residue class $h \pmod{n}$, and its row weight is

$$F_{n,s}(t) = W\left(\frac{t-h}{X}\right) V\left(\log \frac{n^2}{t-h} - s\right). \quad (28)$$

Choose $0 < \alpha < \beta$ with $\text{supp } W \subset [\alpha, \beta]$. On the support of $F_{n,s}$,

$$\begin{aligned} F'_{n,s}(t) &= \frac{1}{X} W'\left(\frac{t-h}{X}\right) V\left(\log \frac{n^2}{t-h} - s\right) \\ &\quad - \frac{1}{t-h} W\left(\frac{t-h}{X}\right) V'\left(\log \frac{n^2}{t-h} - s\right). \end{aligned} \quad (29)$$

Since $t - h \in [\alpha X, \beta X]$, it follows that

$$\|F_{n,s}\|_\infty + \text{Var}(F_{n,s}) \ll_{W,V} 1 \quad (30)$$

uniformly in both n and s .

Stieltjes partial summation therefore bounds the row discrepancy by a constant times

$$\max_{y \ll_{h,W} X} \left| \sum_{\substack{t \leq y \\ t \equiv h \pmod{n}}} \Lambda(t) - \frac{y}{\varphi(n)} \right|.$$

Sum this quantity over $n \leq D$ and apply the maximal Bombieri–Vinogradov theorem [2, 7, 10]. The outer coefficient causes no difficulty. If E_n denotes the nonnegative maximal row discrepancy above, then

$$\sum_{n \leq D} |\Lambda(n) - 1| E_n \leq (1 + \log D) \sum_{n \leq D} E_n.$$

Requesting three additional powers of $\log X$ in Bombieri–Vinogradov absorbs this factor and gives the required total error.

The continuous row main term is exactly

$$\frac{1}{\varphi(n)} \int_{\mathbb{R}} F_{n,s}(t) dt = \frac{X}{\varphi(n)} \int_0^\infty W(u) V\left(\log \frac{n^2}{Xu} - s\right) du.$$

Finally, if $(n, h) > 1$ and $\Lambda(nm + h) \neq 0$, then $nm + h = p^k$ for a prime $p \mid h$. Only $O_{h,W}(1)$ such powers lie in the fixed-ratio target interval. Summing $\Lambda(n) + 1$ over the divisors $n \mid p^k - h$ gives X^ε for every $\varepsilon > 0$, uniformly in s because V is bounded. These rows are absorbed by the stated error. $\square$

Remark 3.2 (The center really may move). If u lies in the support of W , a row n contributes to the main term only when

$$s = 2 \log n - \log X - \log u + O_V(1).$$

Thus the center tracking $n \asymp N$ is approximately $s = \log(N^2/X)$. A fixed center near zero eventually sees no rows deep in the range $n \leq X^{1/2}(\log X)^{-B}$. Uniformity for all s is what makes [theorem 3.1](#) a useful localization statement.

Corollary 3.3 (A compact nonzero-Mellin band on the Type-I side). *Fix $A, T > 0$. There is $B = B(A, h, W, T) > 0$ such that, uniformly for $|t| \leq T$ and $D \leq X^{1/2}(\log X)^{-B}$,*

$$\begin{aligned} & \sum_{\substack{n \leq D \\ m \geq 1}} (\Lambda(n) - 1) \Lambda(mn + h) W(mn/X) (n/m)^{it} \\ &= X^{1-it} I_W(-it) \sum_{\substack{n \leq D \\ (n,h)=1}} \frac{(\Lambda(n) - 1) n^{2it}}{\varphi(n)} + O_{A,h,W,T}(X(\log X)^{-A}), \end{aligned} \quad (31)$$

where $I_W(z) = \int_0^\infty W(u) u^z du$.

Proof. Use $(n/m)^{it} = n^{2it} (mn)^{-it}$ and repeat the proof of [theorem 3.1](#) with row weight $u^{-it} W(u)$. Its supremum norm and total variation are $O_W(1 + T)$. The finite arithmetic sum in (31) retains all prime-power source rows. $\square$

Remark 3.4 (No complete-tail subtraction at nonzero frequency). At $t = 0$, the complete collapse (6) can be combined with the Type I evaluation and subtracted. At $t \neq 0$ the complete divisor coefficient is

$$r^{-it} \sum_{d|r} (\Lambda(d) - 1) d^{2it},$$

which does not collapse to $\log r - \tau(r)$. Consequently [corollary 3.3](#) cannot be subtracted from (7) to localize the large-factor tail. The unresolved term contains the $m = 1$ fixed-prime-pair layer discussed in [section 6](#).

There is nevertheless an exact scale-center average at every factor range. Fubini gives

$$\int_{\mathbb{R}} \mathcal{A}_{h, \leq D, V}(X; s) ds = \left(\int_{\mathbb{R}} V(v) dv \right) \sum_{\substack{n \leq D \\ m \geq 1}} (\Lambda(n) - 1) \Lambda(mn + h) W(mn/X). \quad (32)$$

The same identity holds without $n \leq D$. It transfers a zero-mode asymptotic to an integral over all centers, but not to any specified center.

4 Optimal recovery from finite Mellin information

This section is deterministic. It states exactly what follows from a finite list of Mellin samples or jets together with a total-variation bound, before any special property of the von Mangoldt coefficients is used.

4.1 The exact minimax radius

Let $K \subset \mathbb{R}$ be compact and let $\mathcal{M}(K)$ denote the complex finite Borel measures on K , with total-variation norm. For $\phi_1, \dots, \phi_J, V \in C(K)$ define

$$\mathcal{N}\mu = \left(\int_K \phi_j d\mu \right)_{j=1}^J, \quad L_V\mu = \int_K V d\mu. \quad (33)$$

An algorithm is any map $\Phi : \mathbb{C}^J \rightarrow \mathbb{C}$; no linearity or continuity is assumed.

Theorem 4.1 (Exact finite-information minimax identity). *For every $R > 0$,*

$$\begin{aligned} \inf_{\Phi: \mathbb{C}^J \rightarrow \mathbb{C}} \sup_{\|\mu\|_{\text{TV}} \leq R} |L_V\mu - \Phi(\mathcal{N}\mu)| \\ = R \operatorname{dist}_{C(K)}(V, \operatorname{span}\{\phi_1, \dots, \phi_J\}). \end{aligned} \quad (34)$$

In particular, nonlinear reconstruction cannot improve on the best linear reconstruction.

Proof. For $c = (c_j) \in \mathbb{C}^J$, the linear estimator $\Phi_c(y) = \sum_j c_j y_j$ has error at most

$$R \left\| V - \sum_j c_j \phi_j \right\|_{C(K)}.$$

Taking the infimum over c proves the upper bound.

For the reverse inequality, fix any algorithm Φ and restrict to data $\mathcal{N}\mu = 0$. The two measures μ and $-\mu$ produce the same data, and

$$\max\{|L_V\mu - \Phi(0)|, |-L_V\mu - \Phi(0)|\} \geq |L_V\mu|.$$

Put $\mathcal{E}_0 = \operatorname{span}\{\phi_1, \dots, \phi_J\}$. Its annihilator in $\mathcal{M}(K) = C(K)^*$ is $\mathcal{E}_0^\perp = \ker \mathcal{N}$. The Hahn–Banach distance identity gives

$$\sup_{\substack{\mu \in \ker \mathcal{N} \\ \|\mu\|_{\text{TV}} \leq 1}} |L_V\mu| = \operatorname{dist}_{C(K)}(V, \mathcal{E}_0).$$

Multiplication by R completes the proof. This is the scalar, centrally-symmetric form of the classical optimal-recovery principle [9]. $\square$

Take sample points $t_1, \dots, t_q \in \mathbb{R}$ and multiplicities $s_1, \dots, s_q \geq 1$. The observations

$$\hat{\mu}^{(k)}(t_\ell) = \int_K (iu)^k e^{it_\ell u} d\mu(u), \quad 0 \leq k < s_\ell, \quad (35)$$

correspond to the information space

$$\mathcal{E}(\mathbf{t}, \mathbf{s}) = \operatorname{span}\{u^k e^{it_\ell u} : 1 \leq \ell \leq q, 0 \leq k < s_\ell\}. \quad (36)$$

Consequently [theorem 4.1](#) becomes

$$\inf_{\Phi} \sup_{\|\mu\|_{\text{TV}} \leq R} \left| L_V\mu - \Phi((\hat{\mu}^{(k)}(t_\ell))_{\ell,k}) \right| = R \operatorname{dist}_{C(K)}(V, \mathcal{E}(\mathbf{t}, \mathbf{s})). \quad (37)$$

If only the jets $0, \dots, K_0$ at $t = 0$ are known, the recovery space is the polynomial space $\mathcal{P}_{K_0}$.

Proposition 4.2 (Exact noise–resolution tradeoff). *Suppose the observed coordinates are $y_j = \int \phi_j d\mu + e_j$ with $\|\mu\|_{\text{TV}} \leq R$, $|e_j| \leq \varepsilon_j$, and every $\varepsilon_j > 0$. Then the worst-case minimax error is*

$$\inf_{c \in \mathbb{C}^J} \left\{ R \left\| V - \sum_{j=1}^J c_j \phi_j \right\|_{\infty} + \sum_{j=1}^J \varepsilon_j |c_j| \right\}. \quad (38)$$

Proof. Apply the proof of [theorem 4.1](#) to the product variable (μ, e) with observation $\mathcal{N}\mu + e$ and target $L_V \mu$. The uncertainty set is the unit ball of

$$\max \left\{ \|\mu\|_{\text{TV}}/R, \max_j |e_j|/\varepsilon_j \right\},$$

whose dual norm is $R\|f\|_{\infty} + \sum_j \varepsilon_j |c_j|$. The distance from $(V, 0)$ to the range of the adjoint observation map is exactly (38). $\square$

4.2 A fixed finite data set becomes asymptotically trivial

The log-scale range of factor pairs with $mn \asymp X$ has length $2 \log X + O_W(1)$. On an expanding range, every fixed finite set of frequencies and jet orders eventually performs no better, in the minimax sense, than the constant midpoint approximation.

Theorem 4.3 (The fixed-data midpoint limit). *Let $V \in C_c(\mathbb{R})$ satisfy $0 \leq V \leq M$ and $\max V = M > 0$. Fix a finite set of real frequencies and jet multiplicities, independent of L , and let $\mathcal{E}$ be the associated space (36). Assume $1 \in \mathcal{E}$. Then*

$$\lim_{L \rightarrow \infty} \text{dist}_{C([-L, L])}(V, \mathcal{E}) = \frac{M}{2}. \quad (39)$$

Proof. The constant $M/2$ gives the upper bound. Suppose, to the contrary, that there are $\eta < M/2$, numbers $L_j \rightarrow \infty$, and $q_j \in \mathcal{E}$ with $\|V - q_j\|_{C([-L_j, L_j])} \leq \eta$. On any fixed nondegenerate interval the restriction map on the finite-dimensional space $\mathcal{E}$ is injective, so norm equivalence makes the coefficient vectors of q_j bounded. A subsequence converges in $\mathcal{E}$ to an exponential polynomial q , and

$$\|V - q\|_{C(\mathbb{R})} \leq \eta. \quad (40)$$

In particular q is bounded on $\mathbb{R}$. A bounded exponential polynomial $\sum_{\ell} P_{\ell}(u) e^{it_{\ell} u}$ with distinct real t_{ℓ} has all P_{ℓ} constant: divide by the largest power of u and use the positive mean square of a nonzero trigonometric polynomial. Hence q is a finite trigonometric polynomial. Such a function is uniformly almost periodic; there are arbitrarily large τ for which

$$\sup_{u \in \mathbb{R}} |q(u + \tau) - q(u)| < \delta \quad (41)$$

for any $\delta > 0$; see [3].

Choose u_0 with $V(u_0) = M$ and then choose an almost period τ so large that $u_0 + \tau \notin \text{supp } V$. From (40) and (41),

$$M - \eta \leq |q(u_0)| \leq |q(u_0 + \tau)| + \delta \leq \eta + \delta.$$

Letting $\delta \downarrow 0$ contradicts $\eta < M/2$. $\square$

Remark 4.4 (Fixed means fixed). The frequencies and jet multiplicities in [theorem 4.3](#) do not vary with L . If they are allowed to depend on X , the exact distance formula (37) remains true at each scale, but the limit (39) does not follow from this argument.

4.3 How many zero-frequency jets are needed?

The next result gives the correct information-level order for an interior window. It does not say that the required arithmetic derivatives can be estimated.

Theorem 4.5 (Interior jet complexity). *Let $0 < w \leq L/2$, let $0 \leq V \leq 1$ on $[-L, L]$, and suppose $V(0) = 1$ and $V(w) = 0$. If a polynomial p of degree at most K satisfies*

$$\|V - p\|_{C([-L, L])} \leq \varepsilon < \frac{1}{2}, \quad (42)$$

then

$$K \geq \frac{\sqrt{3}}{2} \frac{1 - 2\varepsilon}{1 + \varepsilon} \frac{L}{w}. \quad (43)$$

Conversely, if V is Lipschitz with Lipschitz constant $O(1/w)$, Jackson's theorem gives a polynomial of degree K with

$$\|V - p_K\|_{C([-L, L])} \ll \min \left\{ 1, \frac{L}{Kw} \right\}. \quad (44)$$

Proof. The mean-value theorem and (42) give a $\xi \in (0, w)$ such that

$$|p'(\xi)| \geq \frac{1 - 2\varepsilon}{w}.$$

Bernstein's interior inequality on $[-L, L]$ gives

$$|p'(\xi)| \leq \frac{K}{\sqrt{L^2 - \xi^2}} \|p\|_\infty \leq \frac{2K(1 + \varepsilon)}{\sqrt{3}L}.$$

This proves (43). After rescaling $[-L, L]$ to $[-1, 1]$, Jackson's inequality bounds the best degree- K error by a constant times the modulus of continuity at scale $1/K$, which is $O(L/(Kw))$ under the stated Lipschitz hypothesis; see [8]. $\square$

For factor scales, one may take $L = \log X + O_W(1)$. Thus a fixed-width interior window requires $K \asymp \log X$ zero-frequency jets for constant accuracy at the deterministic recovery level. Whether arithmetic can control that many jets is a separate and substantially harder question.

4.4 What a continuum band would provide

In this subsection assume additionally that $V \in C_c^k(\mathbb{R})$ for some $k \geq 2$. Let $\mathcal{F}V(t) = \int_{\mathbb{R}} V(u) e^{-itu} du$ and $\widehat{\mu}(t) = \int_{\mathbb{R}} e^{itu} d\mu(u)$. Fourier inversion gives

$$\int_{\mathbb{R}} V d\mu = \frac{1}{2\pi} \int_{\mathbb{R}} \mathcal{F}V(t) \widehat{\mu}(t) dt. \quad (45)$$

If only $|t| \leq T$ is controlled, the deterministic truncation error is

$$\leq \frac{\|\mu\|_{\text{TV}}}{2\pi} \int_{|t| > T} |\mathcal{F}V(t)| dt \ll_k \|\mu\|_{\text{TV}} \|V^{(k)}\|_1 T^{1-k} \quad (46)$$

for $k \geq 2$. Also, for $A_V(s) = \int V(u - s) d\mu(u)$, Plancherel yields

$$\int_{\mathbb{R}} |A_V(s)|^2 ds = \frac{1}{2\pi} \int_{\mathbb{R}} |\mathcal{F}V(t)|^2 |\widehat{\mu}(t)|^2 dt. \quad (47)$$

Thus a genuine continuum-band estimate can imply averaged scale localization. Isolated finite samples cannot substitute for such an estimate without additional regularity information.

5 Divisor-geometric null profiles and an exact certificate

The minimax identity is abstract. We now show that its null directions occur on exact divisor grids, rather than only in a continuum model.

5.1 Prescribed finite samples and jets

Theorem 5.1 (Geometric Mellin-null profile). *Let $t_1, \dots, t_q \in \mathbb{R}$ and $s_1, \dots, s_q \geq 1$, and put $D = \sum_{\ell=1}^q s_\ell$. Fix a prime p , set $\Delta = 2 \log p$, and write*

$$Q(z) = \prod_{\ell=1}^q (z - e^{it_\ell \Delta})^{s_\ell} = \sum_{j=0}^D c_j z^j. \quad (48)$$

On the factor pairs

$$r = p^D, \quad n_j = p^j, \quad m_j = p^{D-j}, \quad 0 \leq j \leq D, \quad (49)$$

the scale nodes are

$$u_j = \log(n_j/m_j) = (-D + 2j) \log p. \quad (50)$$

The nonzero complex profile $c = (c_0, \dots, c_D)$ satisfies

$$\left. \frac{d^k}{dt^k} \sum_{j=0}^D c_j e^{itu_j} \right|_{t=t_\ell} = 0, \quad 1 \leq \ell \leq q, \quad 0 \leq k < s_\ell. \quad (51)$$

Proof. The exponential polynomial of the profile factors exactly as

$$\begin{aligned} F_c(t) &= \sum_{j=0}^D c_j e^{it(-D+2j) \log p} \\ &= e^{-itD \log p} Q(e^{it\Delta}). \end{aligned} \quad (52)$$

At $t = t_\ell$, the last factor has a zero of order s_ℓ , proving (51). $\square$

If the sample multiset is invariant under $t \mapsto -t$, with equal multiplicities, then Q has real coefficients. If it also contains $t = 0$, then $\sum_j c_j = Q(1) = 0$. Writing $c = c^+ - c^-$, the positive and negative parts have the same mass. After normalization they give two probability profiles on the same divisor nodes with identical prescribed samples and jets. Their masses in a window V differ whenever $\sum_j c_j V(u_j) \neq 0$.

Corollary 5.2 (Zero jets are finite differences). *For jets of orders $0, \dots, K$ at $t = 0$, take*

$$Q(z) = (z - 1)^{K+1}, \quad c_j = (-1)^{K+1-j} \binom{K+1}{j}. \quad (53)$$

Then all scale moments through order K vanish, while the response to a window is its $(K+1)$ st finite difference on the divisor grid.

Remark 5.3 (Arithmetic support versus arithmetic coefficients). The nodes (49) are genuine factor pairs of one integer, but the profile c is constructed adversarially. It is not the restriction of $\Lambda(n) - 1$. Hence [theorem 5.1](#) proves a limitation of finite-data recovery over coefficient classes; it is not a counterexample for the actual prime residual.

5.2 The row $81 + 2 = 83$

We give a fully exact certificate whose target is prime. Take

$$h = 2, \quad r = 3^4 = 81, \quad r + h = 83. \quad (54)$$

The five factor pairs and normalized scale nodes are

j	0	1	2	3	4
(n_j, m_j)	(1, 81)	(3, 27)	(9, 9)	(27, 3)	(81, 1)
$x_j = u_j / \log 3$	-4	-2	0	2	4

(55)

Define two probability profiles

$$\mu_+ = \frac{1}{8}(1, 0, 6, 0, 1), \quad \mu_- = \frac{1}{8}(0, 4, 0, 4, 0). \quad (56)$$

Proposition 5.4 (Exact four-jet ambiguity). *The profiles in (56) have the same moments of orders 0, 1, 2, 3:*

$$(M_0, M_1, M_2, M_3) = (1, 0, 4, 0). \quad (57)$$

Consequently their Mellin transforms have identical derivatives through order three at $t = 0$. If $V \in C_c(\mathbb{R})$ satisfies

$$V(0) = 1, \quad \text{supp } V \subset (-\log 3, \log 3), \quad (58)$$

then

$$\int V d\mu_+ = \frac{3}{4}, \quad \int V d\mu_- = 0. \quad (59)$$

On the prime-target row (54), the corresponding formal target-weighted gap is $(3/4) \log 83$.

Proof. Direct substitution at the five integer nodes in (55) gives (57). Equality of derivatives follows because the k th derivative multiplies the k th moment by $(i \log 3)^k$. The support condition in (58) makes the window-value vector $v = (0, 0, 1, 0, 0)$, so (59) follows from the central masses in (56). Finally $\Lambda(83) = \log 83$ is common to every factor pair of $r = 81$. $\square$

The certificate also has an exact minimax value.

Proposition 5.5 (Sharp five-node minimax radius). *On the nodes $x = (-4, -2, 0, 2, 4)$, let $v = (0, 0, 1, 0, 0)$ and let $\mathcal{P}_3$ be the restrictions of polynomials of degree at most three. Then*

$$\text{dist}_{\ell^\infty}(v, \mathcal{P}_3) = \frac{3}{8}. \quad (60)$$

An optimal polynomial is

$$q(x) = \frac{5}{8} - \frac{x^2}{16}. \quad (61)$$

Proof. The vector

$$c = (1, -4, 6, -4, 1) \quad (62)$$

annihilates $1, x, x^2, x^3$ on the five nodes. Since $\|c\|_1 = 16$ and $\langle c, v \rangle = 6$, duality gives the lower bound $6/16 = 3/8$. At the five nodes the residual $v - q$ for (61) alternates between $3/8$ and $-3/8$, proving the matching upper bound. $\square$

The two probability profiles in (56) are the normalized positive and negative parts of (62). Any estimator receiving their common four-jet data must therefore make error at least half of the $3/4$ window gap on one of them. This realizes the radius in (60) without floating-point computation.

6 Taylor jets and the exact selector boundary

The zero Mellin mode treated in TPC-9 is distinguished by the convolution identity $\Lambda * \mathbf{1} = \log$ [11]. This section computes what happens at the first two jets away from that mode. The calculation is exact at each finite X : it does not supply an asymptotic estimate for either jet. We then identify precisely what additional Mellin information would recover the single factor layer $m = 1$. The resulting identity is a reduction to the fixed-shift Hardy–Littlewood problem, not an evaluation of that problem.

Throughout this section, $h \in \mathbb{Z} \setminus \{0\}$ is fixed, $W \in C_c^1((0, \infty))$ is real-valued, and $\Lambda(q)$ is extended by zero to $q \leq 0$. Recall the factor-scale Mellin polynomial

$$\mathcal{A}_h(X; W; t) = \sum_{m, n \geq 1} (\Lambda(n) - 1) \Lambda(mn + h) W(mn/X) \left(\frac{n}{m}\right)^{it}. \quad (63)$$

The sum is finite for every X , so it is an entire function of t .

6.1 The first three divisor-scale coefficients

Put $r = mn$ and use $d = n$ as the divisor variable. The polynomial on the factor fibre above r is

$$B_r(t) := \sum_{d|r} (\Lambda(d) - 1) \exp(it(2 \log d - \log r)). \quad (64)$$

Consequently,

$$\mathcal{A}_h(X; W; t) = \sum_{r \geq 1} \Lambda(r + h) W(r/X) B_r(t). \quad (65)$$

For $k \geq 0$, define

$$D_k(r) := i^{-k} B_r^{(k)}(0) = \sum_{d|r} (\Lambda(d) - 1) (2 \log d - \log r)^k, \quad (66)$$

$$T_j(r) := \sum_{d|r} \Lambda(d) (\log d)^j, \quad Q_k(r) := \sum_{d|r} (2 \log d - \log r)^k. \quad (67)$$

Termwise differentiation in (65) gives

$$i^{-k} \frac{\partial^k}{\partial t^k} \mathcal{A}_h(X; W; t) \Big|_{t=0} = \sum_{r \geq 1} \Lambda(r + h) W(r/X) D_k(r). \quad (68)$$

Proposition 6.1 (Exact zeroth, first, and second jets). *Let $L = \log r$. Then*

$$D_0(r) = L - \tau(r), \quad (69)$$

$$D_1(r) = 2T_1(r) - L^2 = L^2 - 2(\Lambda * \Lambda * \mathbf{1})(r), \quad (70)$$

$$\begin{aligned} D_2(r) &= 4T_2(r) - 4LT_1(r) + L^3 - Q_2(r) \\ &= L^3 - Q_2(r) - 4((\mathfrak{D}\Lambda) * \Lambda * \mathbf{1})(r), \end{aligned} \quad (71)$$

where $\mathfrak{D}f(q) := f(q) \log q$. If $r = \prod_p p^{a_p}$, then

$$Q_2(r) = \frac{\tau(r)}{3} \sum_{p^{a_p} || r} a_p (a_p + 2) (\log p)^2. \quad (72)$$

Proof. Expanding the power in (66) gives, for every $k \geq 0$,

$$D_k(r) = \sum_{j=0}^k \binom{k}{j} 2^j (-L)^{k-j} T_j(r) - Q_k(r). \quad (73)$$

The standard identities

$$T_0(r) = \sum_{d|r} \Lambda(d) = L, \quad Q_0(r) = \tau(r),$$

prove (69). Divisor pairing $d \leftrightarrow r/d$ shows that $Q_1(r) = 0$. Substituting $k = 1$ and $k = 2$ in (73) therefore yields the first expressions in (70) and (71).

It remains to prove the convolution forms. The operator $\mathfrak{D}$ is a derivation for Dirichlet convolution:

$$\mathfrak{D}(f * g) = (\mathfrak{D}f) * g + f * (\mathfrak{D}g). \quad (74)$$

Write $\ell(q) = \log q$. Since $\Lambda * \mathbf{1} = \ell$, applying (74) once gives

$$\ell^2 = (\mathfrak{D}\Lambda) * \mathbf{1} + \Lambda * \ell,$$

where powers of ℓ denote pointwise powers. On the other hand,

$$T_1 = (\mathfrak{D}\Lambda) * \mathbf{1}, \quad \Lambda * \ell = \Lambda * \Lambda * \mathbf{1}.$$

Thus $2T_1 - \ell^2 = \ell^2 - 2\Lambda * \Lambda * \mathbf{1}$, proving the second identity in (70).

For the second jet it is convenient to introduce

$$C_2(f, g)(r) := \sum_{d|r} f(d)g(r/d)(\log d - \log(r/d))^2.$$

Then $D_2 = C_2(\Lambda, \mathbf{1}) - C_2(\mathbf{1}, \mathbf{1})$ and $C_2(\mathbf{1}, \mathbf{1}) = Q_2$. Expanding the square gives

$$C_2(\Lambda, \mathbf{1}) = (\mathfrak{D}^2\Lambda) * \mathbf{1} - 2(\mathfrak{D}\Lambda) * \ell + \Lambda * \ell^2.$$

Applying $\mathfrak{D}^2$ to $\Lambda * \mathbf{1} = \ell$ gives

$$\ell^3 = (\mathfrak{D}^2\Lambda) * \mathbf{1} + 2(\mathfrak{D}\Lambda) * \ell + \Lambda * \ell^2.$$

Subtracting the middle terms proves

$$C_2(\Lambda, \mathbf{1}) = \ell^3 - 4(\mathfrak{D}\Lambda) * \ell = \ell^3 - 4(\mathfrak{D}\Lambda) * \Lambda * \mathbf{1},$$

which is (71).

Finally, write a divisor of r as $d = \prod_p p^{j_p}$ with $0 \leq j_p \leq a_p$. Then

$$2 \log d - \log r = \sum_p (2j_p - a_p) \log p.$$

All mixed terms vanish after summation over the divisor box, since

$$\sum_{j=0}^a (2j - a) = 0,$$

while

$$\frac{1}{a+1} \sum_{j=0}^a (2j - a)^2 = \frac{a(a+2)}{3}.$$

Multiplying by the number $\tau(r)$ of divisors proves (72). $\square$

The convolution $\Lambda * \Lambda * \mathbf{1}$ in the first derivative and the differentiated two- Λ convolution in the second derivative are not Titchmarsh divisor weights. Thus [proposition 6.1](#) identifies the first arithmetic objects that must be estimated away from the zero mode; it does not estimate them or assert that cancellation in their shifted prime-target sums is impossible.

6.2 Even symmetrization does not remove the second interface

Define the reversible part

$$\mathcal{A}_h^{\text{ev}}(X; W; t) := \frac{1}{2} \{ \mathcal{A}_h(X; W; t) + \mathcal{A}_h(X; W; -t) \}. \quad (75)$$

Swapping m and n in the second summand gives the exact identity

$$\mathcal{A}_h^{\text{ev}}(X; W; t) = \frac{1}{2} \sum_{m, n \geq 1} (\Lambda(n) + \Lambda(m) - 2) \Lambda(mn + h) W(mn/X) \left(\frac{n}{m} \right)^{it}. \quad (76)$$

The function is even, so all its odd derivatives at zero vanish. This removes D_1 by construction, but

$$\left. \frac{\partial^2}{\partial t^2} \mathcal{A}_h^{\text{ev}}(X; W; t) \right|_{t=0} = - \sum_{r \geq 1} \Lambda(r + h) W(r/X) D_2(r). \quad (77)$$

In particular, the two- Λ term in (71) remains. Algebraic parity in the Mellin variable is therefore different from the sieve parity issue: killing odd Taylor coefficients does not evaluate the first surviving even coefficient.

6.3 Product demodulation and an exact $m = 1$ selector

The variable t in (63) records only the ratio n/m . To separate a factor endpoint without conflating ratio and product, introduce the two-Mellin polynomial

$$\mathcal{M}_h(X; W; t, s) := \sum_{m, n \geq 1} (\Lambda(n) - 1) \Lambda(mn + h) W(mn/X) \left(\frac{n}{m} \right)^{it} \left(\frac{mn}{X} \right)^{is}. \quad (78)$$

Along the demodulating line $s = -t$, set

$$\begin{aligned} E_{h,X}(t) &:= X^{-it} \mathcal{M}_h(X; W; t, -t) \\ &= \sum_{m, n \geq 1} (\Lambda(n) - 1) \Lambda(mn + h) W(mn/X) m^{-2it}. \end{aligned} \quad (79)$$

Equivalently, if $W_t(u) = u^{-it} W(u)$, then

$$E_{h,X}(t) = X^{-it} \mathcal{A}_h(X; W_t; t). \quad (80)$$

Thus control of $E_{h,X}$ is a two-variable, or coupled-weight, statement; it is not supplied by a theorem for the single value $\mathcal{A}_h(X; W; 0)$.

For $a > 0$, define

$$\kappa_a(t) := \frac{1}{\pi a} \left(\frac{\sin(at)}{t} \right)^2, \quad \kappa_a(0) := \frac{a}{\pi}. \quad (81)$$

With the Fourier convention $\hat{\kappa}_a(\xi) = \int_{\mathbb{R}} \kappa_a(t) e^{-it\xi} dt$, one has

$$\hat{\kappa}_a(\xi) = \left(1 - \frac{|\xi|}{2a} \right)_+. \quad (82)$$

Proposition 6.2 (Exact endpoint selector). *For every $X > 0$ and every $0 < a \leq \log 2$,*

$$\int_{\mathbb{R}} \kappa_a(t) E_{h,X}(t) dt = \sum_{n \geq 1} (\Lambda(n) - 1) \Lambda(n + h) W(n/X). \quad (83)$$

Proof. The transform formula (82) follows by writing

$$\frac{\sin(at)}{t} = \frac{1}{2} \int_{-a}^a e^{itu} du$$

and convolving the two interval indicators. With the convention used here, the Fourier transform of a product is $(2\pi)^{-1}$ times the convolution of the transforms; this gives precisely the normalization in (82). In particular, $\kappa_a \geq 0$, $\kappa_a \in L^1(\mathbb{R})$, and $\int \kappa_a = \widehat{\kappa}_a(0) = 1$.

The sum defining $E_{h,X}$ is finite, so termwise integration is valid. Using (82), its integral is

$$\int_{\mathbb{R}} \kappa_a(t) E_{h,X}(t) dt = \sum_{m,n \geq 1} (\Lambda(n) - 1) \Lambda(mn + h) W(mn/X) \widehat{\kappa}_a(2 \log m).$$

For $m = 1$ the final factor is one. If $m \geq 2$, then $2 \log m \geq 2 \log 2 \geq 2a$, and the final factor is zero. Exactly the $m = 1$ layer remains, proving (83). $\square$

The kernel in [proposition 6.2](#) uses all real Mellin frequencies; its t^{-2} tails are essential to the compact support of $\widehat{\kappa}_a$. The proposition is therefore not a bounded-frequency localization theorem.

6.4 The Hardy–Littlewood equivalence boundary

Let h be an admissible fixed even shift, and let

$$\mathfrak{S}(h) := \prod_p \frac{1 - \nu_h(p)/p}{(1 - 1/p)^2}, \quad \nu_h(p) := \#\{b \bmod p : b(b+h) \equiv 0 \pmod{p}\}. \quad (84)$$

The weighted fixed-shift Hardy–Littlewood assertion is

$$\sum_{n \geq 1} \Lambda(n) \Lambda(n+h) W(n/X) = \mathfrak{S}(h) X \int_0^\infty W(u) du + o_{h,W}(X). \quad (85)$$

The general fixed-shift assertion represented by (85) remains open for every prescribed admissible shift, including $h = 2$ [6, 7].

Corollary 6.3 (Exact equivalence, not an asymptotic evaluation). *For the fixed shift h and the fixed test function W above, fix $0 < a \leq \log 2$. The single weighted assertion (85) is equivalent to*

$$\int_{\mathbb{R}} \kappa_a(t) E_{h,X}(t) dt = (\mathfrak{S}(h) - 1) X \int_0^\infty W(u) du + o_{h,W}(X). \quad (86)$$

Proof. By the shifted prime number theorem and partial summation,

$$\sum_{n \geq 1} \Lambda(n+h) W(n/X) = X \int_0^\infty W(u) du + o_{h,W}(X); \quad (87)$$

see, for example, Iwaniec and Kowalski [7]. Subtracting (87) from (85), and then applying the exact identity (83), gives (86). Reversing the subtraction proves the converse. $\square$

Remark 6.4 (Strict interpretation). Neither side of (86) is evaluated in this paper. The equivalence is for the current fixed test function W ; it should not be read as replacing the full family of Hardy–Littlewood statements by one possibly degenerate weight. The exact selector says that an asymptotic theorem strong enough to control the all-frequency, coupled-weight expression on the left would already contain the fixed-shift Hardy–Littlewood problem. It does not turn the zero-mode estimate of TPC–9 into such control. Likewise, the identities for D_1 and D_2 expose parity-sensitive shifted convolution terms but prove neither their cancellation nor an impossibility theorem. No fixed-shift prime-pair asymptotic, twin-prime lower bound, new Type II estimate, or breach of the sieve parity barrier is claimed.

7 Conclusion and the next analytic target

The factor-scale coordinate separates four levels of information that are easy to conflate.

- (i) The zero Mellin mode admits the complete divisor collapse $\sum_{d|r} (\Lambda(d) - 1) = \log r - \tau(r)$ and hence a fixed-shift Titchmarsh evaluation.
- (ii) Coarse two-sided localization is still possible: after deleting every row with $m \leq Y$ or $n \leq Y$, the leading aggregate survives whenever $\log Y = o(\log X)$. This is the bulk theorem [theorem 2.5](#).
- (iii) Fine localization is available on the Type I side. Maximal Bombieri–Vinogradov evaluates an arbitrary fixed smooth window in $\log(n/m)$, uniformly in its moving center, as long as the modulus factor satisfies $n \leq X^{1/2}(\log X)^{-B}$.
- (iv) Fine localization of the complete long tail cannot be read from a fixed finite list of Mellin samples and a norm bound. The exact error is a best exponential-polynomial approximation distance, and divisor-power fibres realize explicit blind directions.

In spectral language, $u = \log(n/m)$ is the coordinate of the multiplicative dilation flow and e^{itu} is its character. The zero mode is the scale-invariant aggregate, derivatives at zero are generator moments, and a local window requires a frequency superposition. This is a precise use of spectral coordinates; it does not identify zeta zeros, prime gaps, or any other arithmetic data with dynamical eigenvalues.

The first Taylor derivatives explain why complete nonzero-Mellin control is harder than the zero mode. Already D_1 and D_2 contain shifted two-von-Mangoldt convolution coefficients. Even symmetrization removes the former but leaves the latter. At the strongest endpoint, the exact sinc-squared filter recovers the $m = 1$ fixed-prime-pair layer, showing that an all-frequency theorem of the predicted strength would be equivalent to the weighted Hardy–Littlewood assertion.

The most concrete next analytic target is therefore intermediate between the results proved here and that endpoint. One may seek a continuum-band mean-square estimate for the complete large-factor transform,

$$\int_{|t| \leq T} \left| \sum_{\substack{n > D \\ m \geq 1}} (\Lambda(n) - 1) \Lambda(mn + h) W(mn/X) (n/m)^{it} - \mathcal{M}_{h,D}(X; t) \right|^2 dt, \quad (88)$$

with an explicit model $\mathcal{M}_{h,D}$ and a range $T = T(X)$ that grows. Together with a compatible inverse transform for $\mathcal{M}_{h,D}$ and control of the weighted tail $|t| > T$ —or, alternatively, for a band-limited window—[\(47\)](#) would turn such an estimate into an L^2 theorem over scale centers. Averaging the shift h , enlarging the window width, or restricting the frequency band are honest weaker targets if the fixed-shift form [\(88\)](#) proves inaccessible.

Nothing in the present argument supplies signed Type II cancellation, pointwise localization of the complete tail, a fixed-shift prime-pair asymptotic, a twin-prime lower bound, a new level of distribution, or a parity-barrier breakthrough. The advance is instead a sharp map: two regions of rigorous localization, an exact finite-information boundary, and an explicit identification of the arithmetic object that the next layer must control.

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